A chart is an argument

TC1037 · Algorithmic Thinking · Session 13 · Plots with `matplotlib`

Dr. Leopoldo Rodríguez Salazar

Tuesday 10 November 2026

Tecnológico de Monterrey · Campus Querétaro · Group 510

The hour, in order

When	What	
17:35–17:55	**Quick exam 5**	Lists. Twenty minutes, Respondus and Trinket. **On paper if you are on a Mac** — come get a copy.
After	**Plots**	Everything you need for Friday's lab and for two of the five functions in your final program.

Nothing new is examined today that was not on Friday's slides: the accumulator, `sum`/`max`/`min`, `sorted()` against `.sort()`, a row as a list, a table as a list of rows, `tabla[i][j]`.

Due dates — Friday and Friday

When	What	What counts as done
Fri 13 Nov 23:59	**Deliverable 2** `entrega2.docx`	Four to six *preguntas detonadoras* about *your* table. In pairs, one file. The exact shape is on the last slides today.
Fri 16:30	**Lab 7** lists and charts	The same lab as every TC1037 group. Three chart functions plus a menu. Today is the rehearsal.
Rolling	**Deliverable 1 fixes**	If I sent you a list of fixes, upload the corrected file. **No penalty** for fixing it.

Deliverable 1 — what came back, and a correction

Twenty-eight of thirty handed something in, and **thirteen of the fourteen teams wrote the note** — in a second sheet, further along the first sheet, or a separate PDF.

Nobody had duplicate rows. That part the whole group got right.

Each team has a message from me with its own list of fixes. Those lists stand.

Where I was wrong.

I told seven teams their documentation was missing. It was not. I read the first sheet of the first file and stopped — and your notes were in the second sheet, in the columns beside the table, or in a second file.

That was my mistake, not yours. Corrections are going out today. Nothing you submitted loses a point for it.

One team wrote the best note in the group — source, link, licence, every column explained — and picked a table with no numbers in it. One team handed in the cleanest data I have seen in this course and no note at all.

The four faults — and what each one does to your program

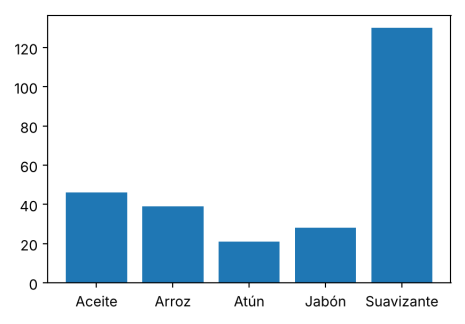
In the file	In the program, next week
Blank rows or columns left in	`len()` counts them. Your mean is divided by the wrong number. **No error message.**
A **constant** column — the same value in all 40 rows	Nothing to group by, and its average is that value. It is not data, it is a label.
An **identifier** used as a variable — receipt number, time stamp, URL, address	Unique in every row. You cannot count categories in it and its mean means nothing.
Numbers stored as **text**	`"10" < "9"` is `True`. Comparisons go alphabetical and the maximum is wrong.

Your first chart — five lines

```
import matplotlib.pyplot as plt

productos = ["Aceite", "Arroz", "
    Atun",
              "Jabon", "Suavizante"
    ]
precios   = [46, 39, 21, 28, 130]

plt.bar(productos, precios)
plt.show()
```



And it says nothing.

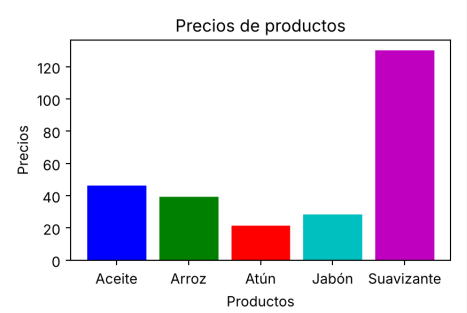
Two lists of the **same length**. The first is what you read on the x axis, the second is how tall each bar is. Position 0 with position 0.

The same chart, with the argument put back in

```
plt.title("Precios de productos")
plt.xlabel("Productos")
plt.ylabel("Precios")

barras = plt.bar(productos,
    precios)
barras[0].set_color('b')
barras[1].set_color('g')
barras[2].set_color('r')
barras[3].set_color('c')
barras[4].set_color('m')

plt.show()
```

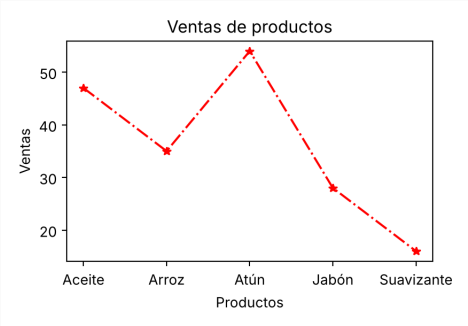


`plt.bar` **returns** the bars. `barras` is a list — one entry per bar — so `barras[0]` is the first bar, exactly like any list you wrote on Friday.

The line chart — and the three-character style string

```
ventas = [47, 35, 54, 28, 16]

plt.title("Ventas de productos")
plt.xlabel("Productos")
plt.ylabel("Ventas")
plt.plot(productos, ventas, "r*-."
    )
plt.show()
```



`"r*-."` is three decisions in one string:
`r` the colour · `*` the marker at each point · `-.` the line style.

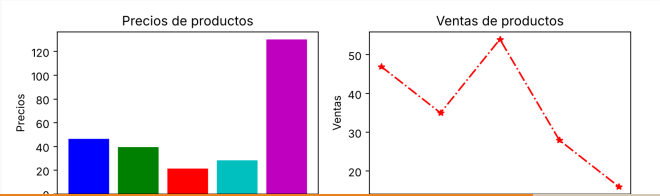
Change it and nothing breaks.

Two charts, one figure — subplot

```
plt.subplot(121)                          # 1 row, 2 columns, drawing in panel 1
plt.title("Precios de productos")
plt.xlabel("Productos"); plt.ylabel("Precios")
plt.bar(productos, precios)

plt.subplot(122)                          # same grid, now panel 2
plt.title("Ventas de productos")
plt.xlabel("Productos"); plt.ylabel("Ventas")
plt.plot(productos, ventas, "r*-.")

plt.show()                                # ONE show, at the very end
```



The shape of the program — you already know it

```
def menu():
    print("1. Grafica de precios")
    print("2. Grafica de ventas")
    print("3. Grafica de precios y ventas")
    print("4. Salir")
def main():
    productos = [...]         # the three lists
    precios   = [...]
    ventas    = [...]
    continua = True
    while continua == True:
        menu()
        opcion = int(input("Opcion: "))
        if opcion == 1:
            precios_productos(productos,
    precios)
        elif opcion == 4:
            continua = False
main()
```

Nothing here is new.

Functions — week 2. `if/elif/else` — week 2. `while` — week 3. Lists — Friday. The only new line is `plt.bar`.

Every `input()` lives in `main`.

The chart functions receive lists and draw. They never ask the user anything.

This is also the skeleton of your **final program** — same menu, same loop, with `estadistica_descriptiva` added.

Names: copy them exactly

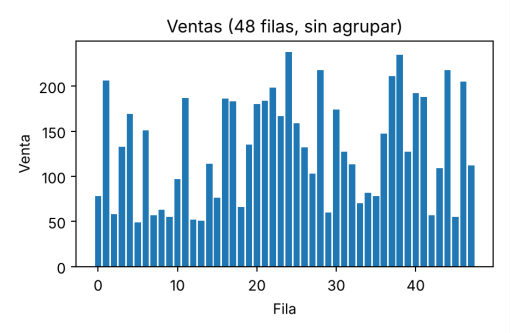
The lab and the project both use **Spanish** function names.

Friday's lab	`precios_productos` · `ventas_productos` · `precios_ventas` · `menu` · `main`
Your final program	`estadistica_descriptiva` · `grafica1` · `grafica2` · `menu` · `main`

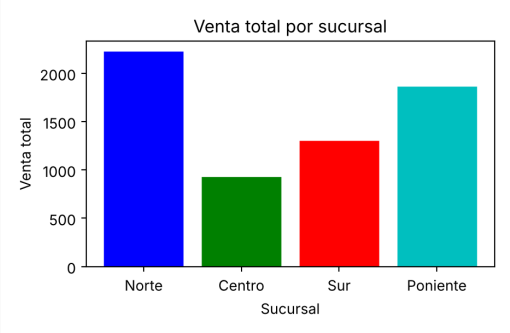
A name that is nearly right is wrong. `graficar1`, `Grafica1` and `grafica_1` are three different functions, and none of them is the one being looked for.

File for Friday: `G_Matricula.py` — your own matrícula, a `.py` file, uploaded to the lab activity.

The trap: you cannot plot your table



Forty-eight rows, one bar each. True, complete — and it answers nothing.



The same forty-eight rows. **Group first, then draw.**

matplotlib needs two short lists. Your table has forty rows. The work in between is the loop you wrote on Friday.

From a table to two lists — Friday's loop, one more time

```
tabla = [["Norte", 120], ["Centro", 80], ["Sur", 95], ["Norte", 60], ["Sur", 40]]

nombres = []        # the x axis: one entry per branch
totales = []        # the y axis: the total for that branch

for fila in tabla:
    sucursal = fila[0]
    venta    = fila[1]
    if sucursal in nombres:
        i = nombres.index(sucursal)
        totales[i] = totales[i] + venta        # the accumulator, again
    else:
        nombres.append(sucursal)
        totales.append(venta)

plt.bar(nombres, totales)
```

`nombres` → `['Norte', 'Centro', 'Sur']` `totales` → `[180, 80, 135]`

Deliverable 2 — due Friday. The exact shape.

Four to six preguntas detonadoras about your own table. One Word file per team.

Each question must carry three things with it:

1. **The question** a manager would actually ask. *Which branch sells most?* Not *what is in the data?*
2. **The column or columns** that answer it — by their real names, as they are spelled in your file.
3. **The statistic or the chart** that answers it. *Total by branch, bar chart.*

And at the end: **who owns which**. Each of you takes two or three, written down.

Marks: 50 for questions answerable *from your columns* · 30 for naming the column and the statistic or chart · 20 for the ownership.

One worked example — and one that fails

Question	Columns	Statistic / chart
Which branch sells most?	`Sucursal`, `Venta`	Total by branch · bar chart
Which product is returned most often?	`Producto`, `Devuelto`	Count by product · bar chart
Did sales grow from January to June?	`Mes`, `Venta`	Total by month · line chart
Is the expensive line also the best seller?	`Precio`, `Venta`	Two panels, one figure
Are our customers satisfied?	—	No column holds this.

The fourth one is `grafica2` in your final program, already chosen. The red one is a good business question and a bad deliverable-2 question.

Asesorías — the offer is open, and it is working

I cancelled sessions two weeks ago without telling anyone. I apologised for that on Friday and the schedule has been clear since.

Slots are being taken. Those who came brought a file and left with it working. That is the whole mechanism — there is no other trick to this course.

Booking link: the Canvas announcement. One click, pick a time.

Bring one specific thing:

- your table, if I sent you fixes and you are not sure what they mean;
- your four questions, before Friday, for ten minutes of sanity checking;
- the chart that will not draw;
- your midterm paper, if the mark still does not make sense to you.

Fifteen minutes on a real problem beats an hour of nodding.

Before Friday

1. **Fix your table** if I sent you a list. It is three lines of work and there is no penalty.
2. **Write the four to six questions** — question, columns, statistic or chart, who owns it. `entrega2.docx`, Friday 23:59.
3. **Run today's five lines** on your own two lists before the lab, so Friday is practice and not a first attempt.
4. **Book an asesoría** if any of the three above is not obvious by Thursday.

A chart that needs you standing next to it is not finished.

Title, both axis labels, and a legend when there is more than one series. Every time.